

Response of American Beech Regeneration to Selection Cutting of Northern Hardwoods in New York

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ABSTRACT. Root sprout and seedling regeneration of American beech was sampled in two maple-dominated northern hardwood stands 13–14 years after single-tree selection cutting. The largest stems had been advance regeneration, and seedlings comprised a substantial and well-distributed portion of these. Among stems established after the cut, root sprouts outnumbered seedlings. Total beech regeneration abundance was correlated with basal area of overstory beech. For long-term beech control, it may be useful to reduce beech seedling populations before selection cutting, and to cut large diameter beech trees.

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In some parts of the Northeast, cutting stimulates dense beech understories that interfere with the regeneration of commercially desirable species (Marquis et al. 1975, Kelty and Nyland 1981, Ostrofsky and McCormack 1986). The problem becomes worse in areas with high white-tailed deer populations because deer preferentially browse species other than beech (Curtis and Rushmore 1958). Although techniques for controlling beech advance regeneration have been tested in even-aged silvicultural systems (Kelty and Nyland 1981, Horsley and Bjorkbom 1983, Ostrofsky and McCormack 1986), methods for uneven-aged systems have not been adequately investigated, in part because of our incomplete knowledge of how beech regeneration develops after partial cutting. We do not know

whether such regeneration is predominantly of sprout or seedling origin, or if most successful regeneration becomes established before (precut) or after (postcut) cutting.

In this paper we report: (1) a comparison of the proportions of four types of beech regeneration (pre- and postcut seedlings and sprouts), 13–14 years after single-tree selection cuttings in two northern hardwood stands; and (2) size and distributional characteristics for each type. We tested whether pre- or postcut stems became proportionally dominant, and whether seedlings or sprouts were more numerous within establishment types. The findings are used to suggest alternatives for controlling beech in the application of uneven-aged systems.

METHODS

Site Descriptions

The study sites are located in New York's Adirondack Mountains (Town of Northville, Saratoga County), and glaciated Allegheny Plateau (Cuyler Hill State Forest, Cortland County). Both are part of a long-term study of growth and yield in northern hardwoods using selection system for uneven-aged management (Nyland et al. 1976, Mader and Nyland 1984). The stands were cut to a residual structure approximating that recommended by Arbogast (1957) in fall 1972 and fall 1973, respectively. Both stands are dominated by sugar maple. Neither had been treated for at least 20 years prior to the selection cutting, except for a small amount of timber stand improvement work at Cuyler Hill. Overstory beech trees were originally more common at Northville than at Cuyler Hill, but were less common at Northville after cutting (Table 1).

The Northville soils are Spodosols and generally coarser in texture than those at Cuyler Hill. The latter site has primarily Inceptisols. Fragipans underlie portions of both stands.

Data Collection and Analysis

Beech stems 5 in. dbh or less were sampled within permanently located, circular milacre plots during summer 1986. There were 188 plots at Cuyler Hill (92 systematically located on a 100-ft grid, supplemented by 24 randomly located clusters of 4), and 159 at Northville (67 systematic, plus 23 random clusters of 4). The randomly located plot clusters were sampled before and after cutting to provide data on total numbers of beech stems at these two times. However, proportions of sprouts and seedlings were not determined until the present study.

In the present study, all beech stems rooted within random and systematic plots were characterized as sprout, seedling, or unknown origin based on excavation of the root collar. Determination of origin was relatively easy for stems up to 1 in. dbh. Of stems classified as unknown in origin, 75% were more than 1 in. dbh. In addition to origin, total height was measured for each stem, and the time of establishment was determined as precut, postcut, or unknown, based on bud scale scar counts. Stems less than 14 or 15 years old were considered postcut at Cuyler Hill and Northville, respectively. Where possible, the exact age of the stem was recorded. Seedlings that had completely died back and resprouted were rare. Thus, little or no error in identifying establishment period can be attributed to seedling sprouts. Whenever a sprout originated within 2 in. of another on the same parent root (i.e., when sprouts were clumped), only the tallest stem was aged and measured.

For milacre plot data, the G statistic with the Williams correction (Sokal and Rohlf 1981) was used to test hypotheses that pre- and postcut stems were equally abundant at each site, and that seedlings and sprouts were equally abundant within each establishment period at each site. Stems of unknown origin and establishment period were not included in the G-tests.

Many stems at Northville were difficult to age because rapid growth had obscured bud scale scars. We therefore conducted a second survey at Northville to investigate beech regeneration age and growth in more detail as follows. At each of 166 systematically located points the nearest beech less than 1.6 in. dbh (the dbh above which all stems were clearly of precut origin), and within 16.4 ft of the point was sampled. For each sampled beech, total height and annual shoot extension (distances between groups

Table 1. Precut and residual stand structure at both study sites (after Nyland et al. 1976).

Dbh class (in.)	Pre-cut		Residual	
	Beech	Other species	Beech	Other species
	[basal area (ft ² /ac)]			
Northville				
1–5	0.4	2.5	0.0	2.5
6–11	3.3	18.8	1.7	15.8
12–17	5.8	28.7	0.0	24.2
18+	6.3	53.3	0.4	22.5
total	15.8	103.3	2.1	65.0
Cuyler Hill				
1–5	2.1	5.0	2.1	5.0
6–11	0.4	19.2	0.4	18.3
12–17	3.8	35.4	2.5	28.3
18+	2.5	42.5	0.0	24.2
total	8.8	102.1	5.0	75.8

of bud scale scars) were measured, the latter only for years where shoot growth could be unambiguously identified. A basal disk was removed to determine stem age in the lab. Lab and field ages were compared to check for bias in field aging.

RESULTS

Density of beech regeneration before and up to 13–14 years after cutting is shown by size class in Figure 1. At Cuyler Hill, the density of larger beech was less after cutting, likely due to breakage during logging. From years 1 to 6 and 6 to 11, density of larger stems increased while that of smaller stems decreased. A new surge in small size classes was evident by 13 years after cutting. At Northville, regeneration density 1 year after cutting was less than one-half the precut level, again likely due to breakage during logging. Density never recovered to precut levels, presumably due to a balance of in- and out-growth. A

drop in density of small size classes was noted by 14 years after the cut.

As of 1986, total number of beech stems per plot was greater at Northville than at Cuyler Hill (Table 2). However, because of high variability in number of stems per milacre (coefficients of variation generally greater than 100%), estimates of beech stem density within 10% of the true mean with a probability greater than 90% are not possible at either site. Although a number of plots at each site had no beech stems (44.6% at Northville; 62.8% at Cuyler Hill), beech were well distributed in each stand. At Northville only 8 of the 166 supplemental sample points did not have a beech less than 1.6 in. dbh within 16.4 ft.

On milacres, stems established after cutting were more abundant than precut stems at Cuyler Hill ($G = 8.04$, $P < 0.005$) while the reverse occurred at Northville ($G = 46.64$, $P < 0.001$) (Fig. 2). Among pre-cut stems,

sprouts and seedlings were equal in abundance at both sites ($G = 0.39$, $P > 0.5$; and $G = 3.51$, $P > 0.05$ for Cuyler Hill and Northville, respectively) while among postcut stems, sprouts were more numerous at both sites ($G = 32.2$, $P < 0.001$; $G = 6.22$, $P < 0.025$) (Fig. 2).

In the supplemental sample at Northville, the same preponderance of precut stems was found ($G = 12.38$, $P < 0.001$) (Fig. 2). Seedlings were proportionally more abundant in the supplemental than in the milacre sample, a result reflecting differences in the way that abundance is estimated in sample points and plots. Sprouts tend to be more clumped than seedlings (Jones and Raynal 1986), and the sample point technique is known to underestimate density if clumping occurs (Mueller-Dombois and Ellenberg 1974).

Comparison of field and lab ages indicated a bias for underestimating age in the field. For example, of 53 stems designated as unknown in establishment period in the field, 44 were actually established prior to cutting according to lab analysis. Similarly, of 56 identified as postcut, 14 were actually precut. Of 46 identified as precut in the field, only 2 were actually of postcut establishment. Thus, at Northville, where many stems were established within a few years of the cut, the actual dominance of precut stems may be greater than shown for milacres in Figure 2.

At 13–14 years after cutting, beech regeneration established before cutting averaged two to eight times the height of stems established since cutting (Table 2). Although mean height within establishment groups was sometimes higher for seedlings than

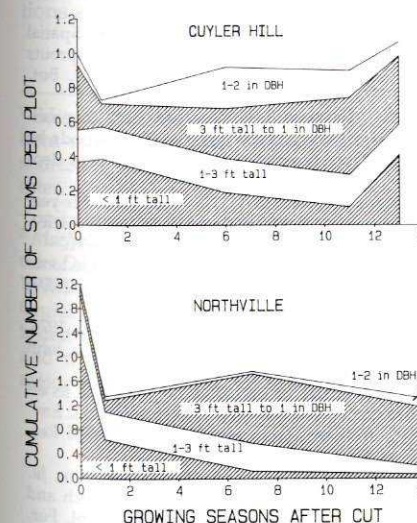


Fig. 1. Size-class distribution of total beech regeneration over time. Data are from 24 randomly located clusters of 4 milacres at each site. Note that the y-axis scale differs between sites.

Table 2. Abundance, distribution, and total height of beech regeneration (stems less than 5 in. dbh) in milacre plots.

Establishment	Origin	Cuyler Hill ^a			Northville ^b		
		No. per plot	% of plots	Height ^c (in.)	No. per plot	% of plots	Height ^c (in.)
Precut	Seed	0.18	15.4	145.2	0.50	25.8	164.3
	Sprout	0.15	10.1	110.8	0.36	20.8	166.6
	Unknown	0.04	3.2	244.6	0.04	4.4	207.8
	Total	0.37			0.90		
Postcut	Seed	0.12	9.6	19.7	0.09	9.4	39.4
	Sprout	0.41	11.7	15.3	0.20	9.4	22.7
	Unknown	0.00	0.0	—	0.00	0.0	—
	Total	0.53			0.29		
Unknown	Seed	0.02	2.1	64.4	0.25	18.9	65.4
	Sprout	0.05	5.3	63.6	0.27	15.7	50.7
	Unknown	0.01	0.5	137.8	0.00	0.0	—
	Total	0.08			0.52		
Total	Seed	0.32	25.5	93.8	0.85	39.6	88.8
	Sprout	0.62	19.7	43.3	0.84	32.7	90.8
	Unknown	0.04	3.7	231.3	0.04	4.4	127.4
	Total	0.98			1.73		

^a Data for 188 milacre plots, 62.8% with no beech.

^b Data for 159 milacre plots, 44.6% with no beech.

^c Within establishment periods (except total) at each site, seedling and sprout heights were significantly different in one case: postcut stems at Northville ($t = 2.26$; $P < 0.05$; $df = 45$).

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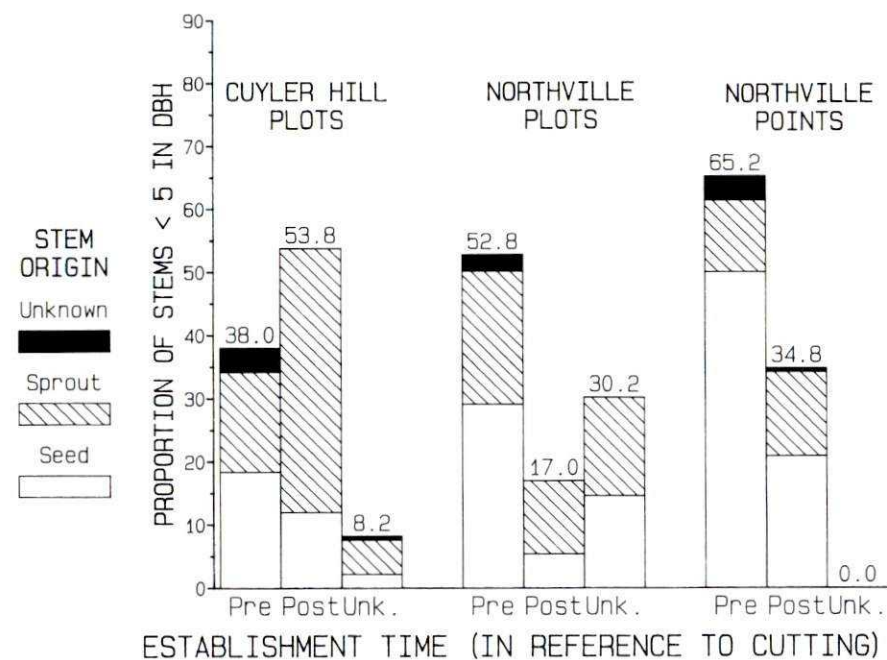


Fig. 2. Proportional distribution of beech regeneration types in 1986.

for sprouts, the difference was significant in only one instance (Table 2). A plot of terminal annual extension for beech in the supplemental sample at Northville shows consistently greater growth among pre-cut seedlings than among sprouts or postcut seedlings (Fig. 3).

DISCUSSION

Since beech was only a minor component in these stands, regeneration of desirable species was sufficiently abundant and vigorous to fulfill long-term structure goals, although a predominance of beech and other undesirable species in the small regeneration size classes was noted 6 years after the cut (Mader and Nyland 1984).

By 13–14 years after cutting, relatively many beech were found in the understory, especially at Northville, where deer browsing is more of a problem. The predominant component of the beech understory at this

time consisted of stems established before cutting. Such stems comprised 38–65% of all beech regeneration and exhibited much greater total height than did postcut stems. Since seedlings formed a large proportion of pre-cut stems and had relatively rapid height growth following cutting, advance seedlings should be viewed as potentially important precursors of postcutting beech understory development.

Stems established after the cut, although shorter than pre-cut regeneration, are nonetheless important because they will serve as advance regeneration that may respond to future cuttings. In general, sprouting was proportionally more important in the post- than in the pre-cut populations. If future selection cuttings continue to stimulate sprouting in this way, sprouts may play an increasing role in future stand structure.

Annual shoot extension measured in this study was less than that of Yetter and Runkle (1986) reported for beech saplings in clearcuts but greater than reported for beech saplings in small canopy gaps. The dramatic drop in annual growth in 1986 (Fig. 3) was due to dieback following a strong, late June frost. The distribution of frost damage was patchy, with some plots showing consistently heavy damage and others showing none. This suggests that chance meteorological events may strongly influence growth patterns, perhaps to an extent that masks differences between sprouts and seedlings.

Abundance of pre- and postcut regeneration was correlated with overstory beech stocking. Northville had twice the beech basal area of Cuyler

Hill before cutting, but only half as much afterwards. Similarly, Northville had somewhat more than twice the pre-cut regeneration seen at Cuyler Hill, but only half as much postcut regeneration. Since most postcut regeneration was of sprout origin, it is possible that retention of overstory beech benefited sprout initiation or survival at Cuyler Hill. Alternatively, the different levels of sprouting seen in the two stands may reflect geographic variability in propensity to sprout, as noted in other regions (Held 1983). More research is needed to clarify relationships between residual beech and sprouting.

IMPLICATIONS

This study demonstrates the importance of advance seedlings in development of beech understories after cutting. In stands where uneven-aged management is planned and high beech density warrants control of advance beech seedling populations should be considered. However, economical methods to do this have not yet been developed. To reduce levels of postcutting sprouting in these stands, the removal or poisoning of as many large beech as possible may prove useful.

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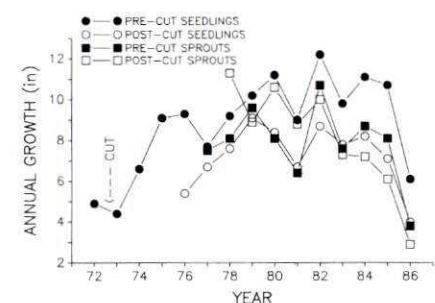


Fig. 3. Mean annual extension growth of beech in the supplemental sample at Northville. Each point represents a mean for 10 or more stems. Coefficients of variation were between 42–92% except in 1986 where they were 88–147%.

FIELD NOTES¹

What Do You Do When There Are "Too Many Trees" in a Sample?

Kim Iles²

When a cruiser encounters "too many" trees in a sample, there are several disadvantages. The extra workload is likely to effect his attitude, his accuracy, or his bookkeeping procedures. You might define "too many" differently than I would, depending on your field conditions. Is there a simple, practical, statistically correct procedure to avoid the problem? Yes there is.

The answer isn't terribly complicated—just split the plot in half and measure one side. This is true for fixed plot sampling as well as variable plot sampling. The cruiser cannot be allowed to make the choice freely, and there must be a way for the check cruiser to verify that he made it "properly." Choosing the side randomly would be nice, but a rule like "odd plot—left side, even to the right" is easier in practice, and shouldn't worry even the most nervous statistician.

There are 3 major advantages to splitting plots, besides increased field efficiency. First, it is simple, easy to do (and check), and usually to the cruiser's advantage. People are inclined to use and perfect systems that are to their advantage. Second, the plot radius doesn't change, so the cruiser is always scanning the same distance for "in" trees. This aids in quality control, speed, and minimizing equipment. Third, it doesn't require changes in the way calculations are made with the computer pro-

gram you are using, it's easy to test the method with your current program, and anyone can understand how and why it works.

Would you like to verify it for yourself, under local conditions, using a computer program you probably don't understand? During your ordinary cruising, when the tree count is more than, say, 10 (with variable plot cruising) record which trees are on which half of the 360° sweep. Place the dividing line along the direction of travel and through the sample point. You can now run the data through your computer program as you normally do, then run it again using the split plot method. How do you get the program to accept a split plot? There are two ways. If the program handles different BAFs, then enter the smaller number of trees but write down twice the BAF you really used. If that isn't allowed, simply enter all the trees *twice*. The program should run without a problem (with a very small probability that the program editor will "object" to several identical trees). Statistics, grade breakdowns, etc. will almost surely be appropriately computed.

There are some practical considerations to bear in mind. The field crew must split the plot properly, so discourage the use of a tree as the boundary marker, and make sure that a full 180° sweep is being made. Keep a long-run tally of the full count vs. the half-plot count. After a few hundred plots you will have a clear indication whether the split-plot count is truly half of the full count, and a comparison of the dbh distributions would be a good confirmation as well. Set a definite upper limit for the splitting action (perhaps 12 trees for prism cruising) and decide that these plots will be split. It is often tempting for the cruiser, if one-half of the plot has only 3 trees, to decide to measure the *whole* plot if that "sparse half" is chosen, but a half plot when the other

"better stocked" side is chosen. This would be biased and must be avoided. You need a definite rule such as "when the total tree count is greater than 12 and each half has at least 3 trees, you will split the plot and choose a half according to (some rule); otherwise you must measure the full plot."

Is there another way to deal with this problem? Why not just change the size of a fixed plot, or the size of the prism you are using? Besides the fact that doing so often introduces complications in the record keeping, computing, or required equipment, you should be aware that doing this is very likely to introduce a bias. The problem is probably not serious when you are reducing the number of trees observed, but it is serious when you are trying to increase the number of trees. Changing the plot size and splitting the plot are different kinds of procedures, and one is biased, while the other is not. The logic and mathematics of this problem have been discussed elsewhere (Iles 1988).

There are people who have tried this idea in the past, mostly on the intuition that it ought to be correct or at least nearly correct. I have never found it discussed in the literature, although semicircular plots have been used for some applications such as in Dilworth and Bell (1986). The method has been adopted by several short courses and institutions in the past few years. There may be other statistically correct methods for approaching this problem, but we have found none that is simpler, as practical and as easy to introduce. What should you do when you have "too few" trees? There is a tremendous temptation to get more trees by moving the plot or some equivalent action. Don't do it. There is nothing wrong with a "zero volume" plot, and trying to avoid it will likely bias your results upward. Accept your lighter workload and put some of that extra time into verifying that the other fieldwork is done more precisely.

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¹ Edited but nonreferred contributions from our readers describing useful ideas, shortcuts, and findings for the field forester.

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